

Circle Packing

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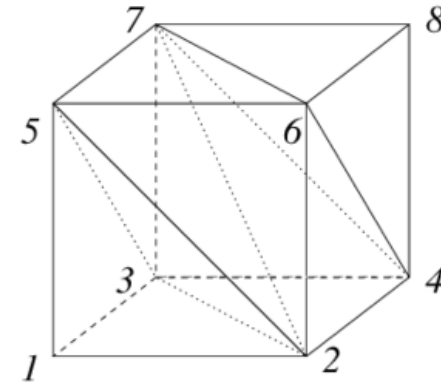
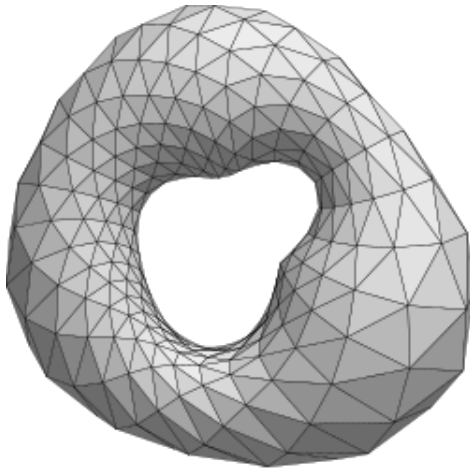
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Triangulated Surfaces

Triangulated Surface

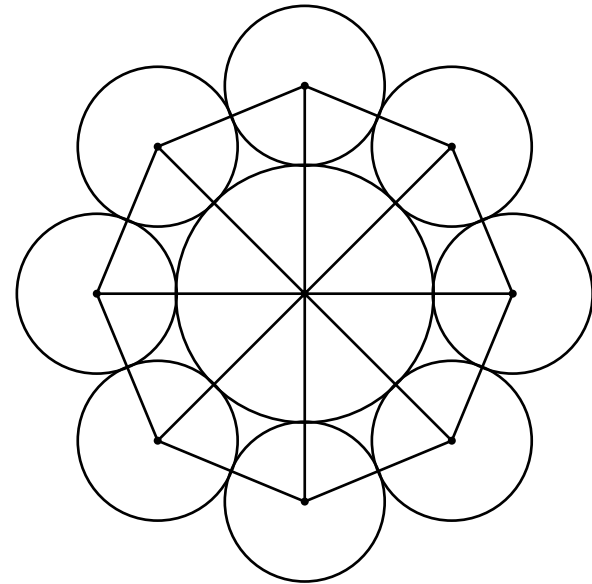
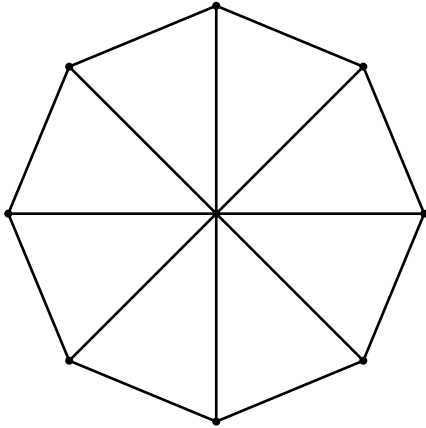
Definition 1

Let $(S, \mathcal{T})$ be a **triangulated surface** where S is a surface and $\mathcal{T}$ is a triangulation obtained by cutting S into triangles and glueing them isometrically on edges such that the result is homeomorphic to S .



Circle Packing from Triangulated Surface

You obtain a circle packing by taking a $(S, \mathcal{T})$, a surface with a triangulation, and applying radius assignments to each vertex, while also enforcing intersection or distance conditions on the circles. The study of circle packing started with the condition that circles of adjacent vertices are tangent.



Radius Function and Edge Lengths

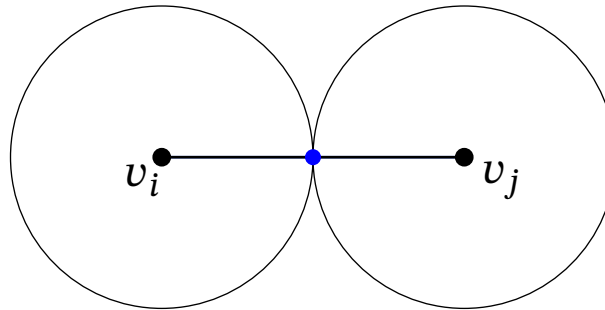
Radius Function

Definition 2

For a triangulated surface $(S, \mathcal{T})$, the **radius function** is a function $r : V \rightarrow (0, \infty)$

A **radius function** induces edge lengths $l : E \rightarrow (0, \infty)$. In the case where all circles are tangent.

$$l_r(v_i v_j) := r(v_i) + r(v_j)$$

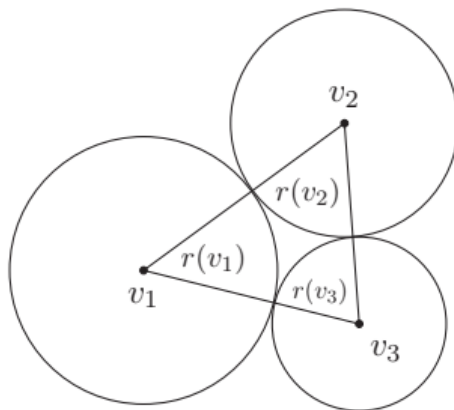


Circle Packing Metrics

Circle Packing Metric

Definition 3

For a triangulated surface $(S, \mathcal{T})$, if a **radius function** r induces edge lengths such that the triangle inequality holds for all triangles in the triangulation then r is **circle packing metric**. In the case of tangent circles, all radius functions are circle packing metrics. [1]



Discrete Curvature

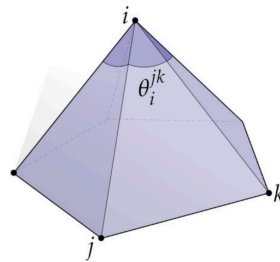
Discrete Curvature

Definition 4

For r a circle packing metric, it induces $K_r : V \rightarrow (-\infty, 2\pi]$ which is the **discrete curvature** where

$$K_r(v_i) := 2\pi - \sum \theta_i^{jk}$$

where the sum ranges over all triangles $\triangle v_i v_j v_k$ with v_i as a vertex and θ_i^{jk} angle at the vertex in the given triangle.



Thurston's Rigidity Theorem for Circle Packings

Thurston's Rigidity Theorem for Circle Packings

Theorem 5

Let $(S, \mathcal{T})$ be a closed triangulated surface. If R and r are circle packing metrics where

$$K_R = K_r$$

Then $R = \lambda r$.

Proven by Thurston [2].

Discrete Schwartz Lemma

Discrete Schwartz-Alfhors-Pick Lemma

Theorem 6

Let $(S, \mathcal{T})$ be a closed triangulated surface where $V(\mathcal{T}) = V_1 \sqcup V_2$. Let R and r be circle packing metrics where

$$K_R|_{V_1} \geq K_r|_{V_1}$$

$$R|_{V_2} \geq r|_{V_2}$$

Then $R \geq r$.

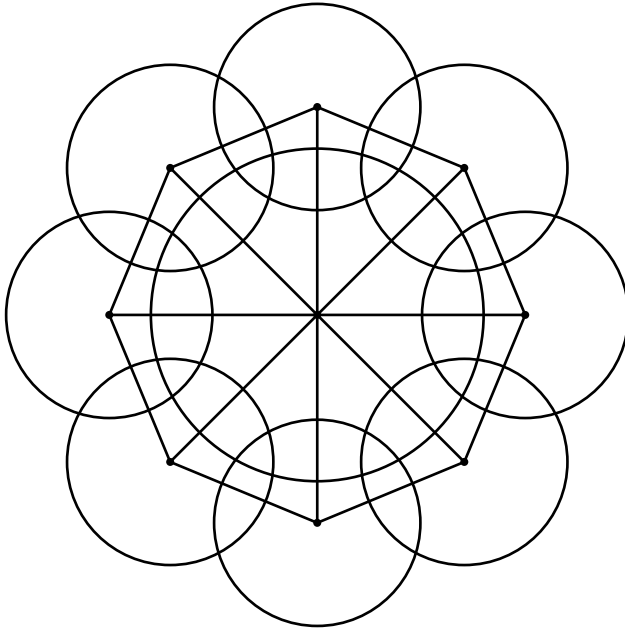
Proven by Beardon and Stephanson in 1991 [3]. We needed a couple key things in the proof:

- The set of circle packing metrics is $\mathbb{R}_{>0}^V$, so its nice and convex.
- The partial derivatives of discrete curvature with respect to radii behave nicely.
- Thurston's Rigidity Theorem

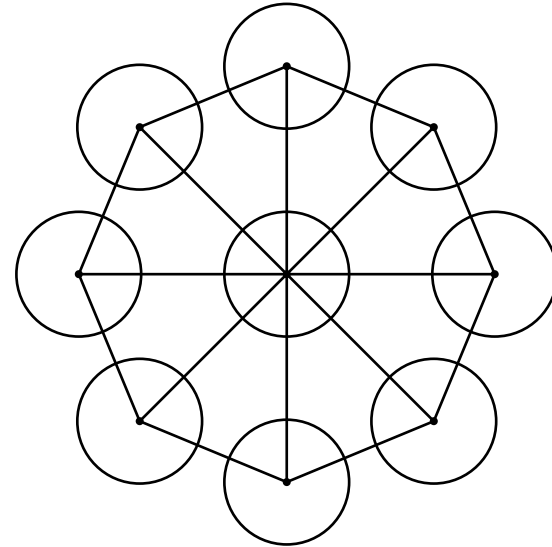
How can we generalize this?

We can loosen the conditions on the circle packings, so the circles of adjacent no longer need to be tangent or even touch.

Intersection Angles



Inversive Distance

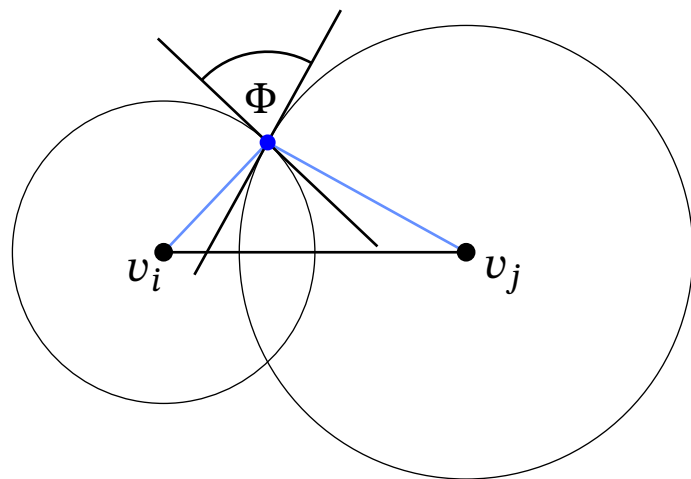


Triangulated Surface With Intersection Angles

Triangulated Surface with Intersection Angles

Definition 7

A **triangulated surface with intersection angles** $(S, \mathcal{T}, \Phi)$ is a triangulated surface with function $\Phi : E(\mathcal{T}) \rightarrow [0, \pi)$. For a edge $v_i v_j$, if you place circles of radii r_i and r_j centered v_i and v_j respectively, then $\Phi(v_i v_j)$ describes the intersection angle between the two circles.



If r is a radius assignment, then the induced edge lengths $l_r : E \rightarrow (0, \infty)$ is given by

$$l_r(v_i v_j) = \sqrt{r(v_i)^2 + r(v_j)^2 + 2r(v_i)r(v_j)\cos(\Phi(v_i v_j))}$$

This follows from a basic geometric calculation using cosine law.

Beardon-Stephenson-Eeuwen theorem

Beardon-Stephenson-Eeuwen theorem

Theorem 8

Let $(S, \mathcal{T}, \Phi)$ be a closed triangulated surface with intersection angles where $V(\mathcal{T}) = V_1 \sqcup V_2$ and $\Phi : E \rightarrow \left[0, \frac{\pi}{2}\right]$. Let R and r be circle packing metrics where

$$K_R|_{V_1} \geq K_r|_{V_1}$$

$$R|_{V_2} \geq r|_{V_2}$$

Then $R \geq r$.

Proven by Jeff Van Eeuwen in 1994 [1].

Discrete Schwartz Lemma for Intersection Angles

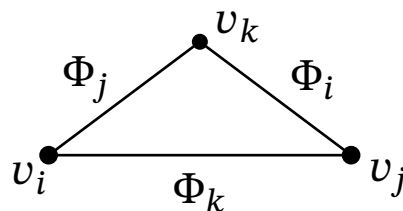
Main Theorem 1

Theorem 9

Let $(S, \mathcal{T}, \Phi)$ be a closed triangulated surface with intersection angles where $V(\mathcal{T}) = V_1 \sqcup V_2$ and $\forall \triangle v_i v_j v_k \in F(\mathcal{T})$

$$\cos(\Phi_a) + \cos(\Phi_b) \cos(\Phi_c) \geq 0$$

for $\{a, b, c\} = \{i, j, k\}$.



Let R and r be circle packing metrics where $K_R|_{V_1} \geq K_r|_{V_1}$ and $R|_{V_2} \geq r|_{V_2}$. Then $R \geq r$.

Triangulated Surface With Inversive Distances

Triangulated Surface with Inversive Distances

Definition 10

A **triangulated surface with Inversive Distances** $(S, \mathcal{T}, I)$ is a triangulated surface with function $I : E(\mathcal{T}) \rightarrow [1, \infty)$. For an edge $v_i v_j$, if you place circles of radii r_i and r_j centered v_i and v_j respectively, then $I(v_i v_j)$ describes the inversive distance between the two circles.

If r is a radius assignment, then the induced edge lengths $l_r : E \rightarrow (0, \infty)$ is given by

$$l_r(v_i v_j) = \sqrt{r(v_i)^2 + r(v_j)^2 + 2r(v_i)r(v_j)I(v_i v_j)}$$

Discrete Schwartz Lemma for Inversive Distances

Discrete Schwartz Lemma for Inversive Distances

Conjecture 11

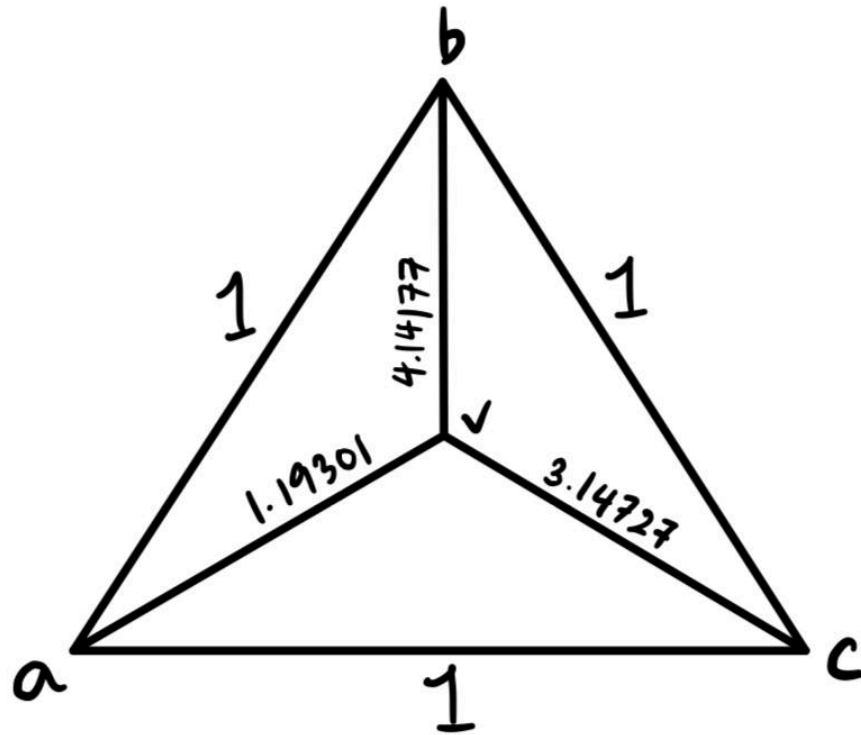
Let $(S, \mathcal{T}, I)$ be a closed triangulated surface with inversive distances where $V(\mathcal{T}) = V_1 \sqcup V_2$.
Let R and r be circle packing metrics where

$$K_R|_{V_1} \geq K_r|_{V_1}$$

$$R|_{V_2} \geq r|_{V_2}$$

Then $R \geq r$.

Counter Example for Discrete Schwartz Lemma for Inversive Distances



Let $V_1 = \{v\}$ and $V_2 = \{a, b, c\}$.

Vertex	Radius $r(v)$	Curvature $k_{r(v)}$
a	1.0	2.49300
b	1.0	4.54271
c	1.0	3.97976
v	1.48297	4.69250

Vertex	Radius $R(v)$	Curvature $k_{R(v)}$
a	1.09193	1.24905
b	2.45799	5.20030
c	2.22062	4.49970
v	1.47981	4.75891

References

- [1] J. V. Eeuwen, “The Discrete Schwarz-Pick Lemma for Overlapping Circles,” *Proceedings of the American Mathematical Society*, vol. 121, no. 4, pp. 1087–1091, 1994, Accessed: Jul. 15, 2025. [Online]. Available: <http://www.jstor.org/stable/2161218>
- [2] W. P. Thurston, *The geometry and topology of three-manifolds: With a preface by Steven P. Kerckhoff*, vol. 27. American Mathematical Society, 2022.
- [3] A. F. Beardon and K. Stephenson, “The Schwarz-Pick lemma for circle packings,” *Illinois Journal of Mathematics*, vol. 35, no. 4, pp. 577–606, 1991.

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